3.2.5 Transition metals (A-level only)

The 3d block contains 10 elements, all of which are metals. Unlike the metals in Groups 1 and 2, the transition metals Ti to Cu form coloured compounds and compounds where the transition metal exists in different oxidation states. Some of these metals are familiar as catalysts. The properties of these elements are studied in this section with opportunities for a wide range of practical investigations.

3.2.5.1 General properties of transition metals (A-level only)

Content	Opportunities for skills development
Transition metal characteristics of elements Ti–Cu arise from an incomplete d sub-level in atoms or ions. The characteristic properties include: • complex formation • formation of coloured ions • variable oxidation state • catalytic activity. A ligand is a molecule or ion that forms a co-ordinate bond with a transition metal by donating a pair of electrons. A complex is a central metal atom or ion surrounded by ligands. Co-ordination number is number of co-ordinate bonds to the central metal atom or ion.	

3.2.5.2 Substitution reactions (A-level only)

Content	Opportunities for skills development
H_2O, NH_3 and Cl^- can act as monodentate ligands. The ligands NH_3 and H_2O are similar in size and are uncharged. Exchange of the ligands NH_3 and H_2O occurs without change of co-ordination number (eg Co^{2+} and Cu^{2+}). Substitution may be incomplete (eg the formation of $[Cu(NH_3)_4(H_2O)_2]^{2+}$). The Cl^- ligand is larger than the uncharged ligands NH_3 and H_2O Exchange of the ligand H_2O by Cl^- can involve a change of co-ordination number (eg Co^{2+}, Cu^{2+} and Fe^{3+}). Ligands can be bidentate (eg $H_2NCH_2CH_2NH_2$ and $C_2O_4{}^{2-}$). Ligands can be multidentate (eg $EDTA^{4-}$). Haem is an iron(II) complex with a multidentate ligand. Oxygen forms a co-ordinate bond to Fe(II) in haemoglobin, enabling oxygen to be transported in the blood. Carbon monoxide is toxic because it replaces oxygen co-ordinately bonded to Fe(II) in haemoglobin. Bidentate and multidentate ligands replace monodentate ligands from complexes. This is called the chelate effect. **Students should be able to:** • explain the chelate effect, in terms of the balance between the entropy and enthalpy change in these reactions.	**AT d and k** **PS 1.2** Students could carry out test-tube reactions of complexes with monodentate, bidentate and multidentate ligands to compare ease of substitution. **AT d and k** **PS 2.2** Students could carry out test-tube reactions of solutions of metal aqua ions with ammonia or concentrated hydrochloric acid.

3.2.5.3 Shapes of complex ions (A-level only)

Content	Opportunities for skills development
Transition metal ions commonly form octahedral complexes with small ligands (eg H_2O and NH_3). Octahedral complexes can display *cis–trans* isomerism (a special case of *E–Z* isomerism) with monodentate ligands and optical isomerism with bidentate ligands. Transition metal ions commonly form tetrahedral complexes with larger ligands (eg Cl^-). Square planar complexes are also formed and can display *cis–trans* isomerism. Cisplatin is the *cis* isomer. Ag^+ forms the linear complex $[Ag(NH_3)_2]^+$ as used in Tollens' reagent.	**MS 4.1 and 4.2** Students understand and draw the shape of complex ions. **MS 4.3** Students understand the origin of *cis–trans* and optical isomerism. Students draw *cis–trans* and optical isomers. Students describe the types of stereoisomerism shown by molecules/complexes.

3.2.5.4 Formation of coloured ions (A-level only)

Content	Opportunities for skills development
Transition metal ions can be identified by their colour. Colour arises when some of the wavelengths of visible light are absorbed and the remaining wavelengths of light are transmitted or reflected. d electrons move from the ground state to an excited state when light is absorbed. The energy difference between the ground state and the excited state of the d electrons is given by: $\Delta E = h\nu = hc/\lambda$ Changes in oxidation state, co-ordination number and ligand alter ΔE and this leads to a change in colour. The absorption of visible light is used in spectroscopy. A simple colorimeter can be used to determine the concentration of coloured ions in solution.	**PS 3.1 and 3.2** Students could determine the concentration of a solution of copper(II) ions by colorimetry. **MS 3.1 and 3.2** Students determine the concentration of a solution from a graph of absorption versus concentration. **AT a, e and k** Students could determine the concentration of a coloured complex ion by colorimetry.

3.2.5.5 Variable oxidation states (A-level only)

Content	Opportunities for skills development
Transition elements show variable oxidation states.	**AT d and k**
Vanadium species in oxidation states IV, III and II are formed by the reduction of vanadate(V) ions by zinc in acidic solution.	**PS 1.2** Students could reduce vanadate(V) with zinc in acidic solution.
The redox potential for a transition metal ion changing from a higher to a lower oxidation state is influenced by pH and by the ligand.	**AT b, d and k** **PS 4.1**
The reduction of $[Ag(NH_3)_2]^+$ (Tollens' reagent) to metallic silver is used to distinguish between aldehydes and ketones.	Students could carry out test-tube reactions of Tollens' reagent to distinguish aldehydes and ketones.
The redox titrations of Fe^{2+} and $C_2O_4^{2-}$ with MnO_4^-	**AT a, d, e and k**
Students should be able to: • perform calculations for these titrations and similar redox reactions.	**PS 2.3, 3.2 and 3.3** Students could carry out redox titrations. Examples include, finding: • the mass of iron in an iron tablet • the percentage of iron in steel • the M_r of hydrated ammonium iron(II) sulfate • the M_r of ethanedioic acid • the concentration of H_2O_2 in hair bleach.

3.2.5.6 Catalysts (A-level only)

Content	Opportunities for skills development
Transition metals and their compounds can act as heterogeneous and homogeneous catalysts. A heterogeneous catalyst is in a different phase from the reactants and the reaction occurs at active sites on the surface. The use of a support medium to maximise the surface area of a heterogeneous catalyst and minimise the cost. V_2O_5 acts as a heterogeneous catalyst in the Contact process. Fe is used as a heterogeneous catalyst in the Haber process. Heterogeneous catalysts can become poisoned by impurities that block the active sites and consequently have reduced efficiency; this has a cost implication. A homogeneous catalyst is in the same phase as the reactants. When catalysts and reactants are in the same phase, the reaction proceeds through an intermediate species. **Students should be able to:** • explain the importance of variable oxidation states in catalysis • explain, with the aid of equations, how V_2O_5 acts as a catalyst in the Contact process • explain, with the aid of equations, how Fe^{2+} ions catalyse the reaction between I^- and $S_2O_8^{2-}$ • explain, with the aid of equations, how Mn^{2+} ions autocatalyse the reaction between $C_2O_4^{2-}$ and MnO_4^-	**AT d and k** **PS 4.1** Students could investigate Mn^{2+} as the autocatalyst in the reaction between ethanedioic acid and acidified potassium manganate(VII).